

Abdullah AL Maruf

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 ResearchGate  Google Scholar  GitHub

I am a Python instructor and research mentor working remotely in India, specializing in Machine Learning, Deep Learning, Natural Language Processing, and Image Processing. My research interests include advancing solutions in computer vision, natural language processing, and low-resource language processing. Previously, I worked as a Java developer at Business Automation Limited. Guided by the motto, "Education and research pave the way for innovation," I aim to bridge theory with real-world applications in AI and software development.

Education

- 2018- 2022  **B.Sc in Computer Science & Engineering (CSE) ,Bangladesh University of Business & Technology (BUBT) .**
Thesis title: *Emotion Detection and Prediction of Human Psychology Using Machine Learning Algorithm .*
Result : 3.44 out of 4 .

Employment History

- 2022 – 2023  **JAVA developer (young professional).** Business Automation Limited, Mirpur DOHS, Dhaka, Bangladesh. As a Java Developer at Business Automation Limited, I worked on services for the Dutch Bangla Bank and government projects, specializing in API development and backend development for platforms like Gas Monkey. I led Cypress automation testing to ensure software quality and gained experience with Spring Boot frameworks.
- 2023 – Continue  **Python Instructor - Remote, India.** As a remote Python instructor in India, I create comprehensive coding courses for an online learning portal and conduct feedback sessions to assist students with any challenges they encounter. This approach helps ensure effective learning and resolution of doubts.
- 2022 – Continue  **Freelance Researcher.** As a freelance researcher, I've collaborated on various projects with international professors, gaining invaluable experience in research methodologies and analytical skills. These unpaid partnerships have expanded my academic network and provided exposure to diverse scholarly perspectives.

Skills

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| Languages |  English, Bengali. |
| Coding |  Java, PHP, Python, C, C++ . |
| Problem Solving |  LeetCode |
| Databases |  MySQL, Oracle . |
| Web Dev |  HTML, CSS, JavaScript, Apache Web Server, Tomcat Web Server. |
| Misc. |  App development (Android) ,Cypress software Testing. |
| Machine Learning |  Supervised Learning (Regression, Classification), Unsupervised Learning (Clustering), Deep Learning (ANN, CNN, RNN, LSTM, Transfer Learning), Ensemble Learning (Bagging, Boosting, Stacking), NLP (Data Preprocessing, Word Embeddings, Transformer), Interpretable Machine Learning (SHAP, LIME, Grad-CAM) |

Skills (continued)

Misc.  Microsoft Office, L^AT_EX, Overleaf, Android Studio, Anaconda, Pycharm, IntelliJ, VS-code, Google Coolab

Area of Research Interest

 Machine Learning, Deep Learning, Natural language processing (LLM, Generative AI, Text Analysis), Image Processing (Classification, Object Detection, Medical Image Analysis), Cyber-security, Cognitive AI, Explainable artificial intelligence (XAI)

Research Publications

Journal Articles

- 1 A. Al Maruf, A. J. Abidin, M. M. Haque, *et al.*, “Hate speech detection in the Bengali language: A comprehensive survey,” *Journal of Big Data*, vol. 11, no. 1, p. 97, 2024, **Scopus Q1**.  DOI: <https://doi.org/10.1186/s40537-024-00956-z>.
- 2 A. A. Maruf, S. H. Fahim, R. Bashar, R. A. Romy, S. I. Chowdhury, and Z. Aung, “Classification of freshwater fish diseases in Bangladesh using a novel ensemble deep learning model: Enhancing accuracy and interpretability,” *IEEE Access*, vol. 12, pp. 96 411–96 435, 2024, **Scopus Q1**.  DOI: 10.1109/ACCESS.2024.3426041.
- 3 A. A. Maruf, F. Khanam, M. M. Haque, Z. M. Jiyad, M. Mridha, and Z. Aung, “Challenges and opportunities of text-based emotion detection: A survey,” *IEEE Access*, vol. 12, pp. 18 416–18 450, 2024, **Scopus Q1**.  DOI: 10.1109/ACCESS.2024.3356357.
- 4 A. A. Maruf, Z. M. Ziyad, M. M. Haque, and F. Khanam, “Emotion detection from text and sentiment analysis of Ukraine Russia war using machine learning technique,” *International Journal of Advanced Computer Science and Applications*, vol. 13, no. 12, 2022, **Scopus Q3**.  DOI: 10.14569/IJACSA.2022.01312101.

Conference Proceedings

- 1 A. J. Abidin, M. K. Siam, A. Al Maruf, R. A. Romy, and Z. Aung, “Leveraging the transfer learning techniques with data augmentation to improve Bangladeshi crop diseases diagnosis,” in *The 7th International Conference on Data Analytics and Cyber Security (DACs)*, **Accepted**, Bihar, India: Springer, 2024.
- 2 A. Al Maruf, A. Golder, A. Al Numan, M. M. Haque, and Z. Aung, “Prediction of heart disease and heart failure using ensemble machine learning models,” in *Intelligent Systems*, S. K. Udgate, S. Sethi, and X.-Z. Gao, Eds., Singapore: Springer Nature Singapore, 2024, pp. 481–492, ISBN: 978-981-99-3932-9.  DOI: https://doi.org/10.1007/978-981-99-3932-9_41.
- 3 M. M. Haque, A. Al Maruf, A. J. Obeid, H. K. Judi, M. H. Bhuiyan, and Z. Aung, “Exploring elderly activity recognition using deep learning through sensor-enabled android app data collection,” in *2024 Third International Conference on Power, Control and Computing Technologies (ICPC2T)*, IEEE, 2024, pp. 582–587.  DOI: 10.1109/ICPC2T60072.2024.10474756.
- 4 S. Hossain, N. N. I. Prova, M. R. Sadik, and A. Al Maruf, “Enhancing crop management: Ensemble machine learning for real-time crop recommendation system from sensor data,” in *2024 International Conference on Smart Systems for Applications in Electrical Sciences (ICSSES)*, IEEE, 2024, pp. 1–6.  DOI: 10.1109/ICSSES62373.2024.10561406.

- 5 Z. M. Jiyad, A. Al Maruf, M. M. Haque, M. S. Gupta, A. Ahad, and Z. Aung, "DDoS attack classification leveraging data balancing and hyperparameter tuning approach using ensemble machine learning with XAI," in *2024 Third International Conference on Power, Control and Computing Technologies (ICPC2T)*, IEEE, 2024, pp. 569–575.  DOI: 10.1109/ICPC2T60072.2024.10475035.
- 6 A. A. Maruf, R. A. Rumey, R. I. Sony, and Z. Aung, "Analysis of factors affecting bangladeshi students' academic performances and their predictions using machine learning with explainable ai techniques," in *27th International Conference on Computer and Information Technology (ICCIT)*, **Accepted, Core: C**, Cox's Bazar, Bangladesh: IEEE, 2024.
- 7 A. L. Roy, M. K. Siam, N. Noor, S. Jahan, and A. Al-Maruf, "Leveraging gene expression data and explainable machine learning for enhanced early detection of type 2 diabetes," in *The 3rd International Conference on Computing Advancements (ICCA)*, **Accepted**, Dhaka, Bangladesh: ACM, 2024.
- 8 A. L. Roy, M. K. Siam, A. Al Maruf, and Z. Aung, "Efficient language identification of Indian languages using MFCCs and deep learning models," in *The 7th International Conference on Data Analytics and Cyber Security (DACS)*, **Accepted**, Bihar, India: Springer, 2024.
- 9 M. R. Sadik, R. I. Sony, N. N. I. Prova, *et al.*, "Computer vision based Bangla sign language recognition using transfer learning," in *2024 Second International Conference on Data Science and Information System (ICDSIS)*, IEEE, 2024, pp. 1–7.  DOI: 10.1109/ICDSIS61070.2024.10594269.
- 10 S. Sahajuddin, A.-A. Ahmed, A. Al Maruf, and F. Khanam, "A novel weighted ensemble strategy using transfer learning for enhanced cervical cancer classification," in *The 3rd International Conference on Computing Advancements (ICCA)*, **Accepted**, Dhaka, Bangladesh, 2024.
- 11 A. Al Maruf, A. Al Numan, M. M. Haque, T. T. Jidney, and Z. Aung, "Ensemble approach to classify spam SMS from Bengali text," in *International Conference on Advances in Computing and Data Sciences*, Springer, 2023, pp. 440–453.  DOI: https://doi.org/10.1007/978-3-031-37940-6_36.
- 12 A. A. Maruf, A. Golder, M. S. Naser, A. J. Abidin, A. A. C. Giti, and Z. Aung, "Development of automatic number plate recognition system of Bangladeshi vehicle using object detection and OCR," in *International Conference on Advances in Data-driven Computing and Intelligent Systems*, Springer, 2023, pp. 331–342.  DOI: https://doi.org/10.1007/978-981-99-9524-0_25.
- 13 A. Al Maruf, M. A.-A. Nayem, M. M. Haque, Z. M. Jiyad, A. M. O. Rashid, and F. Khanam, "A survey on personality prediction," in *Proceedings of the 2nd International Conference on Computing Advancements*, 2022, pp. 407–414.  DOI: <https://doi.org/10.1145/3542954.3543012>.
- 14 A. A. Maruf, M. N. H. Biplob, and F. Khanam, "Covid-19 vaccine sentiment detection and analysis using machine learning technique and NLP," in *International Conference on Machine Intelligence and Emerging Technologies*, Springer, 2022, pp. 401–414.  DOI: https://doi.org/10.1007/978-3-031-34619-4_32.

Articles Under Review

- Journal  TransembleNet: Enhancing Species and Vector Mosquito Classification through Transfer Learning Base Ensemble Model — 1st author. PLOS ONE. **Scopus Q1** (Under third revision).
-  EmoBang: Detecting Emotion From Bengali Texts — 1st author. SN Computer Science. **Scopus Q1** (Under first revision).
-  Fostering Trust and Interpretability in Skin Cancer Classification: A Hybrid Framework of Deep Learning and Machine Learning with Explainable AI — 2nd author. Heliyon. **Scopus Q1** (Under first revision).

Articles Under Review (continued)

- Multi-label Emotion and Sentiment Detection from Bengali Text Using Hybrid Deep Learning Model — 2nd author. Computer Speech & Language. **Scopus Q1** (Under review).

Talks

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| Virtual | <ul style="list-style-type: none">In Proceedings of the 2nd International Conference on Computing Advancements. Dhaka, Bangladesh : Presented our research paper. A Survey on Personality Prediction..International Conference on Machine Learning, IoT and Big Data (ICMIB-2023). Sarang, India: Presented our research paper. Prediction of Heart Disease and Heart Failure Using Ensemble Machine Learning.International Conference on Advances in Computing and Data Sciences (ICACDS-2023). Kolkata, India: Presented our research paper. Ensemble Approach to classify Spam SMS from Bengali text. |
| Physical | <ul style="list-style-type: none">In Proceedings of the International Conference on Machine Intelligence and Emerging Technologies (MIET 2022). Noakhali, Bangladesh: Presented our research paper. Covid-19 Vaccine Sentiment Detection and Analysis Using Machine Learning Technique and NLP |

References

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